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Introduction to Web Clipping

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Overview

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- The Story (what & why)
- The System
- Development Resources
- Demo
- Summary
- Q&A


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The Story (What & Why)
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The Story (What & Why)

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What Is Web Clipping?

- Retrieving a tailored response to a specific request
  - A web clipping = the data returned (response)
  - Meets a single need
  - Doesn't try to meet a myriad of potential needs
    - "Just the facts, ma'am."
  - Like a "clipping" from an article
  - e.g., news, stock quotes, on-line orders, etc.
- HTML 3.2
  - Some extensions and exclusions

The Story (What & Why)

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Why?

- Form factor
  - Hand-held devices are small
    - Screens
    - Memory
    - Battery life
- Time and Money
  - World-wide-wait
  - Unwanted data

The Story (What & Why)

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Web Clipping vs. Web Browsing

- Form factor
  - Web Browsing
    - Web browsing implies an unfocused search
    - Designed to draw “browsers”
      - Appeal to a large audience
    - Designed for abundant resources
      - Large screens, memory
  - Web Clipping
    - Web clipping implies a focused search
    - Designed to deliver your data
    - Designed for moderate resources
      - Small screens, memory, battery life

The Story (What & Why)

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Web Clipping vs. Web Browsing

- Time and Money
  - Web Browsing
    - Faster access / larger pages
    - “Unlimited” access plans support browsing
  - Web Clipping
    - Slower access / smaller pages
    - “Limited” access plans suggest focused retrieval
    - Reduced wait time
    - Pay for what you want

The Story (What & Why)

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Terminology

- Web clipping or clipping
- Web clipping application
  - WCA or PQA
- Web clipping application viewer
  - Viewer or Clipper
- Web clipping proxy server
  - Clipping proxy server or proxy server
- Web clipping application builder
  - Builder

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The System

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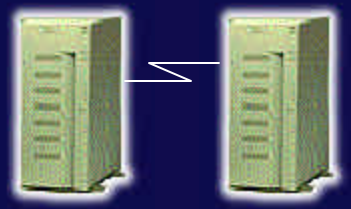
Components

SOFTWARE

HARDWARE







SERVICES

Palm.Net

The System

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Hardware

- Palm VII™ Connected Organizer
  - Wireless Internet access
    - 8 Kbps bursts
    - UDP with RMP (reliable check) plug-in
  - All the features of other Palm™ organizers
    - Date Book, Memo Pad, Address Book, etc.
    - HotSync®
    - IR port
  - 2 Mb RAM and 2 Mb Flash ROM
  - Palm OS® 3.2 (upgradable)
    - A Palm OS 3.5 ROM will be made available for upgrades

The System

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Software (Web Clipping Applications)

- Initiate a wireless Internet transaction
- Stored in the Palm.Net™ category
- Written in HTML
  - 3.2 with some extensions and exclusions
  - gif and jpeg images
- Converted to WCA Format using the WCA Builder
- Rendered on a Palm VII™ device by Clipper
- Like a mini web site
- .pqa file extension

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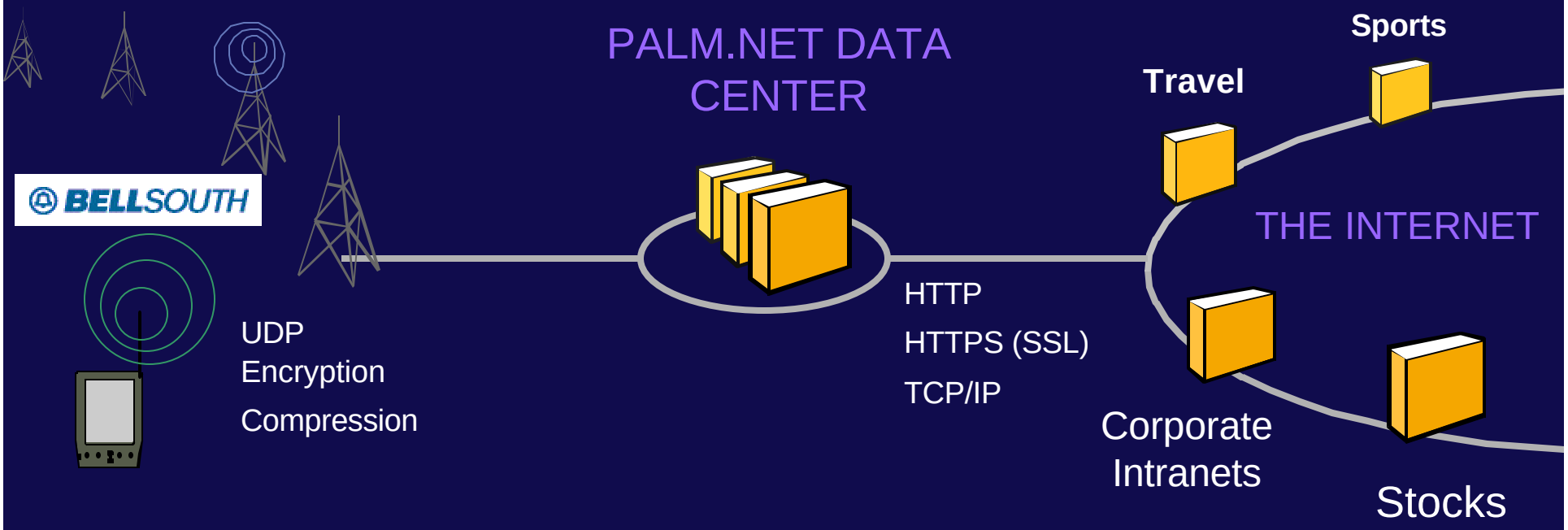
Services

- Palm.Net™ Service
  - 3 service plans
    - Basic (50 Kb / mo.)
    - Expanded (150 Kb/mo.)
    - Volume (300 Kb/mo.)
  - E-mail
    - account_name@palm.net
    - 2 Mb mailbox on Palm.Net server
    - Send and Retrieval initiated from Palm VII™ device
  - 24 x 7
  - Access to account info
  - All Palm VII wireless data flows through Palm.Net service

The System

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Data Flow



The System

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BellSouth Wireless Data

- Mobitex network
  - Developed in 1984 by Eritel
  - International data communication standard
  - Open architecture
  - Managed by Mobitex Operators Association
    - Controls specifications
    - www.mobitex.org

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BellSouth Wireless Data

- Mobitex network
  - 900 MHz channels in U.S.
  - Packet-switched network
    - MPAK (Mobitex packet)
  - Automatic error detection and correction
  - Mobitex Access Number (MAN)
    - Unique to every Palm VII™ device
    - Stored in the radio modem
- Direct line to Palm.Net™ data center

The System

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Palm.Net™ Data Center

- Located in San Jose, California
- Secured site
- Scalable
- Multiple clipping proxy servers
  - Production servers for Palm VII™ devices
  - Development server for Palm OS® Emulator
- Internet access
  - HTTP or HTTPS protocols using TCP/IP
  - T1 connections

The System

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Security: Palm VII™  *data center*

- Standard
  - Frequency agility
  - Bit interleaving and coding
  - Data scrambling as part of Error Control Algorithm
  - Network authentication
  - Message integrity check (MIC)
- Augmented
  - URLs using HTTPS protocol
  - Certicom Elliptic Curve Cryptography technology
    - Public / private keys
    - 163-bit keys = 1024 RSA keys

The System

Security: Data center ⟷ *Internet*

- Standard (data center)
  - Fully redundant firewalls with 1 minute fail-overs
  - Secure Windows NT machines with service pack 5
- Augmented
  - URLs using HTTPS protocol
  - SSL 40-bit or 128-bit connection
    - Microsoft's SChannel.dll
    - Wide range of certificate authorities (CA) recognized
    - Clipping proxy servers do not send certificates
      - Servers requiring a certificate will be refused


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Development Resources

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www.palm.com/devzone/webclipping

- Palm VII™ White Paper
- Web Clipping Developer's Guide
- WCA Builder (or PQA Builder)
- Palm OS® Emulator
- Tutorials
- Example HTML code

Development Resources

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General Palm Development

- Development Zone Knowledge Base
  - oasis.palm.com/devzone/knowledgebase.cfm
  - Searchable
- Developer Assistance
  - http://www.palm.com/devzone/assistance/
  - E-mail forum subscription
  - Newsgroup listings
  - Training info
  - Submit a technical question to Palm Devsupp

What Next?

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- Session 446
  - Learn how web clipping applications are built
- Session 480 Lab
  - Use the tools and build a web clipping application
  - See how easy it really is

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Summary

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Summary

- Web clipping is ideal for the hand-held domain
  - Meets the small form-factor
  - Saves on time and money
  - HTML content
- Palm.Net™
  - Three service plans
  - Mobitex (BellSouth Wireless Data)
  - Security
    - Certicom Elliptic Curve Encryption
    - SSL 40-bit and 128-bit connections


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